Name : .....

Student ID: .....

Signature: .....

The University of New South Wales

School of Electrical Engineering and Telecommunications

Final Examination

Session 1 2015

ELEC3104: Digital Signal Processing

Candidates should attempt ALL questions

1. Reading time: 10 minutes
2. Time allowed: 3 hours.
3. The paper contains 5 questions – answer all questions.
4. The questions are worth 100 marks in total.
5. Print your name, student ID and question number on the front page of each answer book.
6. This exam contributes 70% of the total course assessment.
7. The questions are NOT of equal value. Part marks are as indicated.
8. The examination paper may not be retained by the candidate.
9. This is a closed book exam.
10. The use of UNSW approved calculators is permitted.
11. Transform tables are attached to this exam paper for your use.
12. This paper contains 7 pages apart from this instruction page.

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK.

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Information sheet

Table of the z transforms of useful sequences

Basic z-Transforms	Sequence	z-Transform
1. Unit sample	$\delta[n]$	1
	$\delta[n-k]$	z^{-k}
2. Unit step	$u[n]$	$z/(z-1)$
3. Exponential	$a^n u[(n)]$	$z/(z-a)$
4. Sinusoidal	$\sin[\theta_o n]\, u[n]$	$\dfrac{z \sin\theta_0}{z^2 - 2\cos\theta_0\, z + 1}$
	$\cos[\theta_o n]\, u[n]$	$\dfrac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0\, z + 1}$
5. Unit ramp	$nu[n]$	$\dfrac{z}{(z-1)^2}$

$$1 + z^{-1} + z^{-2} + z^{-3} + \ldots\ldots\ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

z-Transform Properties	Sequence	z-Transform
	$x[n]$	$X(z)$
	$x[n-k]$	$z^{-k} X(z)$
	$a^n x[n]$	$X(z/a)$
	$x[-n]$	$X(z^{-1})$
	$x[n]*y[n]$	$X(z).Y(z)$
	$n\, x[n]$	$-z\,(dX(z)/dz)$

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\cos\theta = \frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right]$$
$$\sin\theta = \frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$$

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Information Sheet

Fourier Transform pair (for discrete-time signal)

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \ldots \qquad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}d\theta$$

θ = *digital frequency*; $-\pi \le \theta \le \pi$

Fourier Transform pair (for continuous-time signal)

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

A **Butterworth filter** of order 'n' is defined by

$$|H(j\omega)|^2 = \frac{1}{1+\left(\frac{\omega}{\omega_c}\right)^{2n}}$$ where ω_c is the cut-off frequency of the filter

n	Butterworth Polynomials
1	$S+1$
2	$s^2+\sqrt{2}s+1$
3	$(s^2+s+1)(s+1)$

Impulse Invariant Method:

$$\frac{1}{s+b} \rightarrow \frac{1}{1-e^{-bT}z^{-1}} \qquad b>0 \quad T\text{-sampling period}$$

Bilinear Transformation:

$$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} \qquad T - sampling\ \ period$$

Probability Density Function (PDF):
The quantisation error (e) is normally assumed to be random and uniformly distributed with zero mean. Therefore, the PDF, $P(e) = 1/\Delta V$, where ΔV is the quantisation step size.

Time Domain	Frequency Domain
$x(n)\cos(\theta_0 n)$	$\frac{1}{2}[X(\theta-\theta_0)+X(\theta+\theta_0)]$
$x(n)\sin(\theta_0 n)$	$\frac{1}{2j}[X(\theta-\theta_0)-X(\theta+\theta_0)]$
$x(n)e^{-j\theta_0 n}$	$X(\theta+\theta_0)$

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Question 1 (20 marks)

(a) Consider a signal, $s(t)$, with a magnitude spectrum satisfying the following equation

$$|S(f)| = \begin{cases} 3 - |f| & 2\text{kHz} < |f| < 3\text{kHz} \\ 0 & \text{otherwise} \end{cases}$$

f – frequency in kHz. The signal $s(t)$ is sampled uniformly with a sampling frequency of 2kHz. Sketch the digital spectrum $|S(\theta)|$.

[5 marks]

(b) A digital communication link carries binary-coded words representing samples of an input signal

$$x(t) = 5\cos(600\pi t) + 4\cos(1800\pi t)$$

The link is operated at 10,000 bits/s and each input sample is quantised into 1024 different voltage levels.

(i) What are the frequencies in the resulting discrete-time signal x[n]?

[2 marks]

(ii) Determine the resulting discrete time signal x[n].

[3 marks]

(c)

(i) A FIR filter has an impulse response as shown in Figure 1.

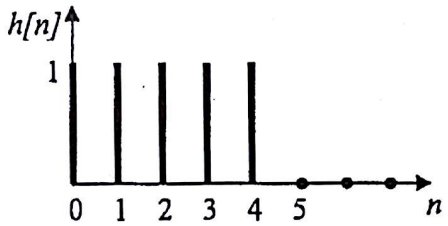


Figure 1

Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$ where θ is the digital frequency.

[4 marks]

(ii) Draw an implementation of this filter using pole and zero configuration.

[4 marks]

(iii)Show that the above filter has a constant group delay. **[2 marks]**

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Question 2 (20 marks)

(a)

(i) Show that the signal to quantisation noise ratio (SQNR) of a linear A/D converter is given by

$$SQNR = 10\log P_s + 10.8 + 20\log L - 20\log R$$

where P_s is the signal power $\left[P_s = \frac{1}{N}\sum_{n=0}^{N-1} x^2[n]\right]$; L is the number of quantisation levels and R is the dynamic range of the input signal, x(t).

[4 marks]

(ii) Using the above equation, show that for a B-bit quantiser, $SQNR = 6.02B + 1.76\ (dB)$ if $x(t) = A\cos(2\pi ft)$. **[3 marks]**

(b) The impulse response , h[n], of a discrete time system is given in Figure 2. Sketch the output signal, $y[n]$, for the given input signal, $x[n]$.

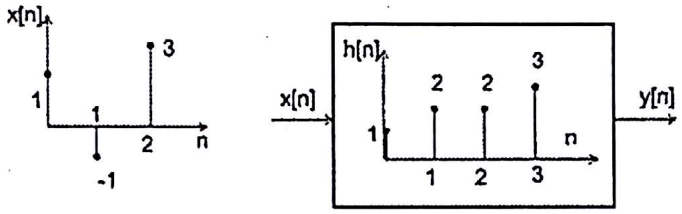


Figure 2

[3 marks]

(c) The following first-order IIR digital filter has been proposed for an application:

$$H(z) = k\frac{1-z^{-1}}{1+b\,z^{-1}} \qquad 0<b<1$$

(i) Determine the value of k in the transfer function $H(z)$, so that the filter has a 0 dB gain at $\theta = \pi$; where θ is the digital frequency.

[2 marks]

(ii) Using the value of k from part (i) above, determine the impulse response of the above digital filter and show that the filter is stable .

[3 marks]

(iii) Show that the 3dB cut-off frequency (θ_c) of the transfer function $H(z)$ from part (i) above, is given by

$$\theta_c = \cos^{-1}\left(\frac{-2b}{1+b^2}\right)$$

[5 marks]

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Question 3 (19 marks)

(a) The pole-zero diagram of a system is shown in Figure 3. The DC gain of the system is $15\left(1-\frac{1}{\sqrt{2}}\right)$.

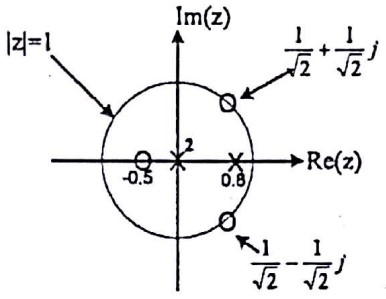


Figure 3

(i) Sketch the approximate magnitude response of the system **[3 marks]**

(ii) Determine the transfer function, $H(z)$, of the system **[4 marks]**

(b) Determine the stability region (triangle) for following causal system.

$$H(z)=\frac{1}{1+b_1z^{-1}+b_2z^{-2}}$$

Your answer should include a sketch and equations of the boundaries of the region. Also indicate the region where complex conjugate poles and real poles exist.

[3 Marks]

(c) Show that the following transfer function can be obtained using a parallel connection of two all-pass filters.

$$H(z)=\frac{14+11z^{-1}}{9+6z^{-1}}$$ **[2 Marks]**

(d) A third order FIR filter has a transfer function G(z) given by

$$G(z)=(9+6z^{-1}-8z^{-2})(2+5z^{-1})$$

From G(z), determine a transfer function of an FIR filter whose magnitude response is identical to that of G(z) and has

(i) A minimum phase response **[2 Marks]**

(ii) A maximum phase response **[2 Marks]**

(e) A first-order digital filter has a transfer function given by:

$$H(z)=\frac{1}{4}\,\frac{1+z^{-1}}{1-0.5z^{-1}}$$

Draw a parallel realisation of $H(z)$. **[3 Marks]**

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Question 4 (23 marks)

(a) It is required to design a digital filter to approximate the analogue filter given by the following transfer function

$$H(s) = \frac{0.1s}{s^2 + 0.1s + 1}$$

Using the bilinear transform show that the resulting digital filter can be written in the form of

$$H(z) = \frac{(a_0 - a_0 z^{-2})}{1 + b_2 z^{-2}}$$

(i) Determine the values of the coefficients a_0 and b_2. The resulting digital filter is to have a resonant frequency of $600Hz$. The sampling frequency is 2400 Hz. **[6 marks]**

(ii) Draw a canonic realisation of $H(z)$. **[3 marks]**

(b) In applications involving the digital modulation of sinusoidal carrier signals, it is necessary to generate sine and cosine signals simultaneously from an oscillator. Consider the system

$$y[n] = a\, y[n-1] + x[n]$$

where $a = e^{j\theta_0}$ and the frequency of oscillation is θ_0. Let the input x[n] be real and the output y[n] is complex. Thus

$$y[n] = y_R[n] + j\, y_I[n].$$

(i) Determine the equations describing a system with one input x[n] and two outputs $y_R[n]$ and $y_I[n]$ only. **[4 Marks]**

(ii) Draw a structure for the realisation of this oscillator. **[3 Marks]**

(iii)Show that if $x[n] = \delta[n]$ then

$$y_R[n] = \cos[n\theta_0].u[n] \quad \text{and} \quad y_I[n] = \sin[n\theta_0].u[n]$$

[3 Marks]

(c) Determine and sketch the phase response of the following filter:

$$y[n] = x[n] - x[n-4]$$

[4 Marks]

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Question 5 (18 marks)

(a) A signal has been sampled at 48 kHz and for a particular application this signal is required to be used at a reduced sampling rate of 10 kHz. Draw a block diagram to show how you would achieve this. Explain how you would efficiently implement a system to achieve this, indicating the type of digital filter you would use (give reasons). **[4 marks]**

(b) Figure 4 shows a modulator scheme where the digital input x[n] is given by

$$x[n] = \frac{1}{2}\delta[n+2] + \delta[n] + \frac{1}{2}\delta[n-2].$$

The carrier frequency of the cosine oscillator $\theta_0 = 3\pi/4$ and the sampling frequency of the input signal $x[n]$ is 8kHz.

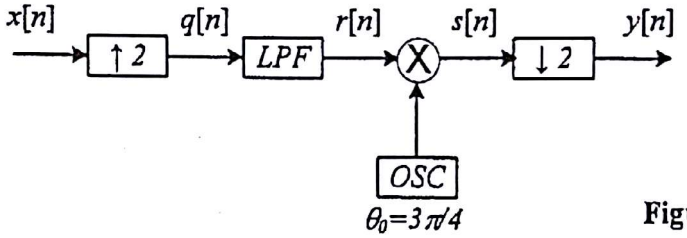


Figure 4

The ideal low pass filter $H_d(\theta)$ (LPF in Figure 4), with linear phase can be characterised in the frequency domain by

$$H_d(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that $Q(\theta) = 1 + cos(4\theta)$ for $-\pi \le \theta \le \pi$ **[2 Marks]**

(ii) If $|X(\theta)|$ is the magnitude spectrum of $x[n]$ sketch the magnitude spectra of $|Q(\theta)|$, $|R(\theta)|$, $|S(\theta)|$ and $|Y(\theta)|$ in separate diagrams. **[8 Marks]**

(c) Show that if x[n] is a real valued signal, the structure shown in Figure 5 has a zero-phase response; where h[n] is the impulse response of the filter H(z) given by,

$$H(z) = (\frac{1}{2} + z^{-1} - \frac{1}{2}z^{-2})$$

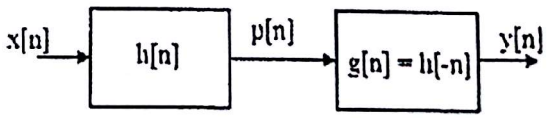


Figure 5

[4 Marks]

End of examination paper.

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